University of St.Gallen

Using Computer Simulation

for Understanding Organizations as Complex Systems

Week 8: How to verify and validate simulation models?

Stephan Aier

From insight to impact.

Structure

Date	Topic	Deliverables	Literature	Responsible
19.02.	Introduction to the course, theory foundations, simulation demos	-	papers 1, 2	Stephan & Marcel
26.02.	How to apply agent-based modelling?	-	book chapter 1	Marcel
05.03.	How to implement simulation models with NetLogo?	-	book chapter 2 (3, 4, 5)	Marcel
12.03.	Independent case work (*no input session*)	-		-
19.03.	Intermediate group presentation and feedback on case/model	Presentation		Stephan & Marcel
26.03.	**How to run simulation experiments and analyse simulation data?**	-	book chapter 6	Stephan
	(*break*)			
16.04.	Intermediate group presentation on implemented model and analysis	Presentation		Stephan & Marcel
23.04.	How to verify and validate simulation models?	-	book chapter 7	Stephan
30.04.	Independent case work (*no input session*)	-		-
07.05.	Final case presentation in groups and reflection of learnings	Presentation, Model, Analysis		Stephan & Marcel
14.05.	Ascension Day (*no input session*)	-		-
21.05.	Oral exams	Oral Exam		Stephan & Marcel

Paper 1: Beese, J., Haki, M. K., Aier, S., & Winter, R. (2019). Simulation-Based Research in Information Systems: Epistemic Implications and a Review of the Status Quo. Business & Information Systems Engineering, 61(4), 503-521.
Paper 2: Haki, K., Beese, J., Aier, S., & Winter, R. (2020). The Evolution of Information Systems Architecture: An Agent-Based Simulation Model. MIS Quarterly, 44(1), 155-184.
Book: Wilensky, U., & Rand, W. (2015). An Introduction to Agent-Based Modeling: Modeling Natural, Social, and Engineered Complex Systems with NetLogo. The MIT Press.

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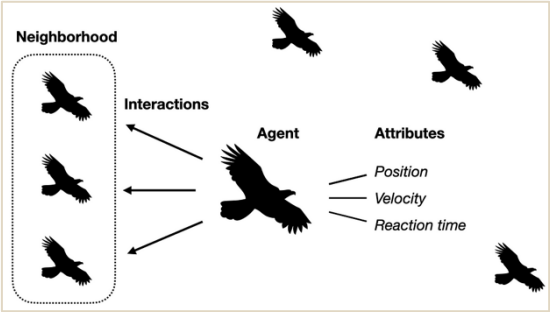
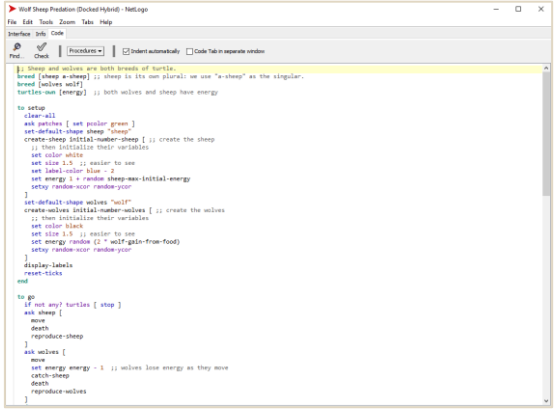
Recap week 6

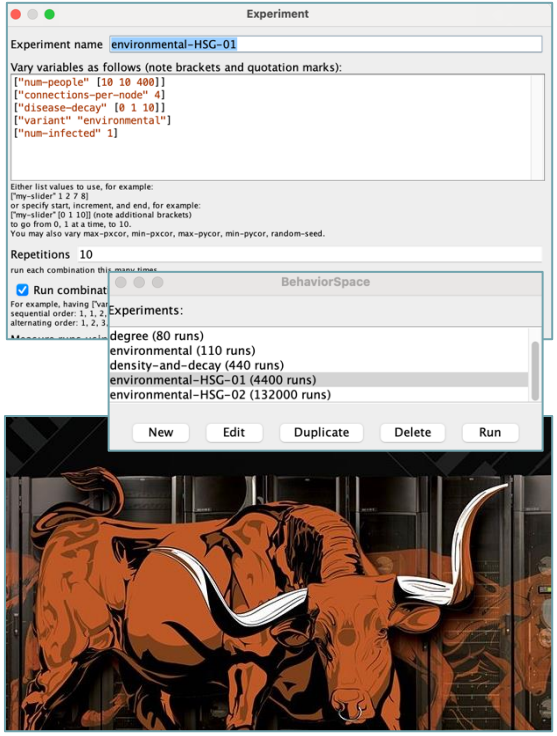
From Models, to Experiments, to Insights

Models

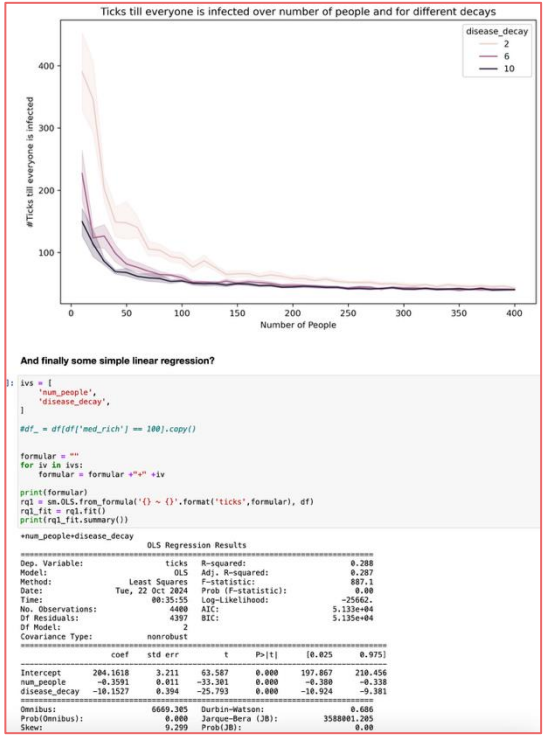




Experiments



Data and Insights



Some observations from 2nd round of group presentations

1. Generally, we have seen **nice ideas, models, implementations** ... some data.

2. Yet, we have seen different **levels of progress and maturity**, from systematically moving forward, to restructuring, to restarting.

3. Depending on your level of progress and maturity you may follow different strategies: **extending/deep diving/polishing** vs. **closing the loop** with a basic model.

4. It is **more important to close the loop, and to discuss and reflect** the results (as well as the process producing those) than to have really new/surprising/disruptive insights.

5. Closing the loop also means: from the real world – to an abstract model – **back to the real world**, i.e., to also discuss the validity of your model and your findings.

Final Presentations: May 7th, 2026

Every group has a slot for presenting the results of their project of **max. 15 minutes** presentation + a short Q&A. Every group member should present.

In the presentation you should cover:

- The problem setting in the real world, including the research question(s)
- Your abstraction of the real world in the conceptual model
- Very short demo of your simulation model
- Your experiments and their results
- The discussion of
  - Your findings and answers to your research questions
  - The validity of your model and thus your findings
  - Your personal take aways

Please find the more detailed time planning / time slot for your presentation here on Canvas.

Oral exams: May 21st, 2026

The exams will take place online via Zoom:
https://unisg.zoom.us/j/65467664866?pwd=ztSQqxneBbNUkA053pmEHxr2TzNaV3.1&from=addon

If you join the zoom session, you will get to a waiting room. We will let you in, once we are ready. Please understand the scheduled times as the earliest possible start of your exam.

Please **make sure you have**

- a stable internet connection
- a quite environment
- a working webcam and audio setup
- your student ID at hand.

We will record the exam sessions for documentation purposes.

Please find the anticipated timeslot for the oral exams on here Canvas

Submission of the simulation model: May 22nd, 2026 (or earlier ;-))

What to submit here via Canvas:

- **NetLogo file** (or equivalent)
  - interface (*Interface* tab)
  - explanatory description (*Info* tab)
  - documented code (*Code* tab)
  - experiment definition (Tools → *BehaviorSpace*)
- documented scripts for **data analysis** (e.g., in Python, Excel)
- **data set** (data generated through simulation experiments) on which the analysis (presentation) is based
- your **presentation slides** from May 5th (while this is not part of the programming grade, it may be helpful for us to understand your simulation)
- *if* significant changes have been made after the presentation, add a short documentation of those changes in the InfoTab

If you need help in the meantime

If you need support for finalizing your project, you can always **reach out to us** via email.

For simple questions we will answer via email.

For more complex questions questions or questions that require discussion, we may schedule a short call.

If you feel your question is in the latter category, please provide 3 possible slots for such a call. We will try to make one of those work.

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We would really like to receive your feedback … really!

https://go.blueja.io/YqHgQBwo-kacUbBBvqRIpQ

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"Your simulation did what you programmed it to do.

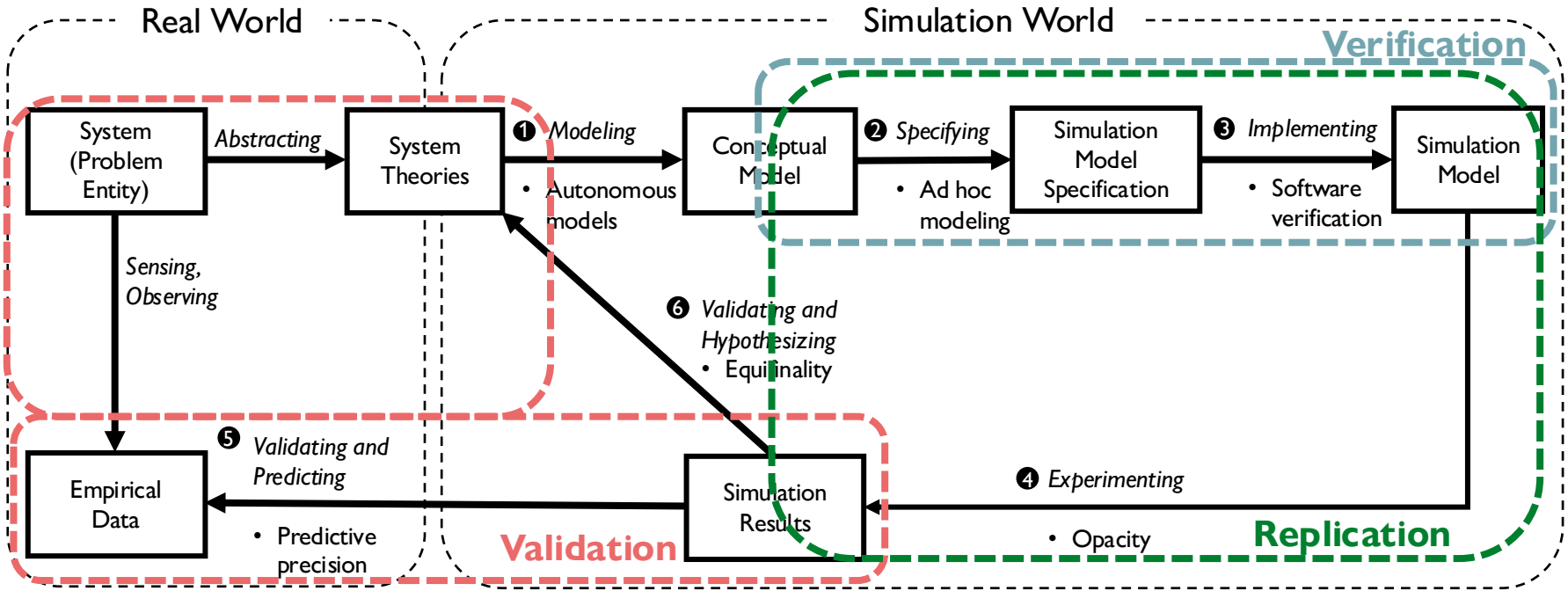
This could be anything!

Where is your research?"

What would you reply?

What can you do to substantiate your claims?

Verification, Validation, Replication



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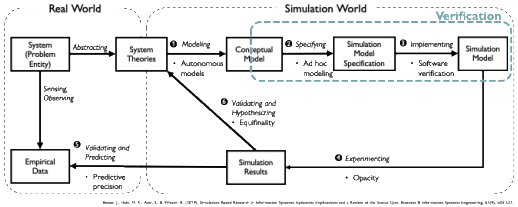
Goal and Benefits



All our models produce some data or results, however, *"it is challenging to determine whether a surprising result is the product of a bug in the code, a miscommunication between the model author and implementer, or an unexpected outcome of low-level rule"* *(Wilensky, U., & Rand, W. (2015). An Introduction to Agent-Based Modeling: Modeling Natural, Social, and Engineered Complex Systems with NetLogo. The MIT Press. p. 325)*

The goal is to make sure the implementer (understood and) **implemented what the model designer specified** (and meant and intended).

> This really helps **understanding causes for unexpected or surprising outcomes**. Because it fosters an understanding and **explication of (all) low-level rules**.

This is less of a problem (NOT no problem) if **model designer and implementer are the same** person/s.

The result is **not binary**!

Approaches

Communication
- Talk to each other
- Make sure you use consistent terminology for the conceptual model and in the source code
- Update terminology in the source code whenever the conceptual model is adapted

Check data and results
- Write unit tests, i.e., every time you setup or run the model, it checks whether the results comply with your theoretical expectations
- Do those checks within your data analysis pipeline

Sensitivity analyses
- Do not just vary the independent variables but also most/all other parameters, e.g., number of ticks, number of agents, …
- Understand how those changes impact the dependent variables (values change too?, in which direction?, does the nature of relations change? distributions?)
- Can you explain those changes? Should the model be calibrated? Do changes of values have implications?

Own experiences

In paper reviews

- We sometimes had reviewers as methods experts
- Associate editors facilitated a “moderated” discussion with reviewers
- Reviewers had full transparency down to source code and data details
- Sometimes reviewers hinted at inconsistencies that resulted from model/code evolutions
- Reviewers were critical to less used tools
- Editors asked to build models as far as possible based on well-published research
- Reviewers questioned “every value”

More and more

- Journals require “full transparency”, i.e., pre-registering conceptual models, simulation models, hypothesis before actually generating/acquiring data
- Journals require full publication of code, models, analysis pipelines, data, …

Implications for our projects

What to do in *our projects*?

1. Make sure you use consistent terminology for the conceptual model and in the source code.
2. Do a sensitivity analysis and discuss the results

Table B2. Tested Parameter Ranges and the Boundary Conditions for the Initialization Parameters

Parameter	*Range Tested*	*Default Value*	*Results*
Time steps	Min. 100 Max. 10,000	1,200	The simulation takes ~150 time steps to stabilize. We therefore ran the simulation for 1,200 time steps, discarded the first 200, and considered the subsequent 1,000 time steps for analysis.
Number of actors	Min. 10 Max. 10,000	100	The observed emergence of outcomes occurs only when the simulated organization has more than about 100 actors.
Number of "?"s in the IT application's feature set	Min. 0 Max. 10	3	This parameter must be balanced with subsequent environmental parameters (i.e., ambiguity and complexity). Having three features that are not fixed provides sufficient room for improvisation without hampering the actor's performance.
Opportunity payoff	Random distribution	$\mu = 30$, $\sigma^2 = 5$	Adopting Davis et al.'s (2009) choice, opportunity payoffs are randomly distributed following a normal distribution with mean $\mu = 30$ and variance $\sigma^2 = 5$.
Opportunity duration	Random distribution	$\mu = 20$, $\sigma^2 = 5$	Adopting Davis et al.'s (2009) choice, opportunities remain for a random number of turns, following a normal distribution with mean $\mu = 20$ and variance $\sigma^2 = 5$.
Initial number of IT applications	Min. 0 Max. 200	50	These parameters are highly stable. An initial number or type of connections that is fundamentally misaligned with the actor's needs will be corrected within 200 time steps in the simulation run. Therefore, we observed no major effects when only analyzing system behavior after the first 200 time steps of a simulation run. Consequently, we chose initial settings that roughly represent the mean values that occur in longer simulation runs and that are well within the observed minimum and maximum ranges.
Initial actor–IT connections per actor	Min. 1 Max. 100	20	

Haki, K., Beese, J., Aier, S., & Winter, R. (2020). The Evolution of Information Systems Architecture: An Agent-Based Simulation Model. MIS Quarterly, 44(1), 155-184.

Table A1. Sensitivity Analysis of Model Parameters

Model Parameter	Description	Model's baseline value	All tested Values	Justification (based on sensitivity analysis)
Agents				
No. of partners	The number of partners is fixed in the entire ecosystem and throughout the simulation runs.	100	25, 50, 100, 200, 500	Based on sensitivity analysis, we observed that outcomes only emerge when there are about 100 partners in the ecosystem. Increasing the number of partners had little effect on the average generativity and profitability. While we did observe an effect on the total generativity and profitability by increasing the number of partners, the results generally followed the same pattern, but with higher values because there were more contributing partners. Additionally, we selected the baseline value considering model efficiency considerations, as a higher number of partners resulted in higher computational penalties for the simulation.
No. of resources per partner	Each partner is endowed with a certain number of resources with random features.	4	4, 5, 6, 7, 8	Sensitivity analysis showed that increasing the number of resources for each partner did not have any meaningful effect on the output variables. In our model, at least four resources are needed to generate one service. Therefore, we endowed each partner with four resources.
No. of platform owner resources	The platform owner offers a defined number of core resources. All services generated on-platform must integrate one core resource.	30	20, 25, 30, 35, 40	The platform owner offers core resources, such as APIs and SDKs, with which partners can integrate their own resources and generate services (e.g., Haki et al., 2024; Schreieck et al., 2021). Sensitivity analysis showed that although a minimum number of platform owner resources is necessary to make the platform attractive, further increasing the number of platform owner resources does not have any effect on the output variables.
Platform owner's revenue share	When a service is generated on-platform, the platform owner receives a defined percentage of the obtained payoff (see "opportunity payoff").	20%	5%, 10%, 20%, 30%, 40%, 50%	Revenue sharing is a means for the platform owner to appropriate value from partners' service innovations on-platform. Based on sensitivity analysis, increasing the platform owner's revenue share has a negative impact on the profitability outcomes of partners. The baseline value for this parameter was selected based on empirical data from other innovation platforms, where the owner's revenue share varies across platforms, such as Apple's 30% on iOS in the B2C context and Salesforce's 15% on AppExchange in the B2B context (Karhu et al., 2018; Staub et al., 2021). Since varying the revenue share only varied partners' profitability, we chose a fixed value within the range of values from empirical data.

...

Haki, K., Tanriverdi, H., Safaei, D., Schmid, M., Aier, S., & Winter, R. (2024). Generativity and Profitability on B2B Innovation Platforms: A Simulation-based Theory Development. MIS Quarterly, 48(2), 583-612. https://doi.org/10.25300/MISQ/2023/17710

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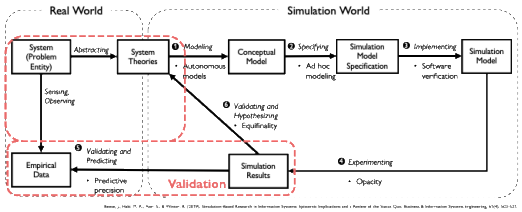


Make sure, our model and its results have *"some correspondence"* to the real world.

"**Microvalidation** is making sure the behaviors and mechanisms encoded into the agents in the model match up with their real-world analogs.

Macrovalidation is the process of ensuring that the aggregate, emergent properties of the model correspond to aggregate properties in the real world.

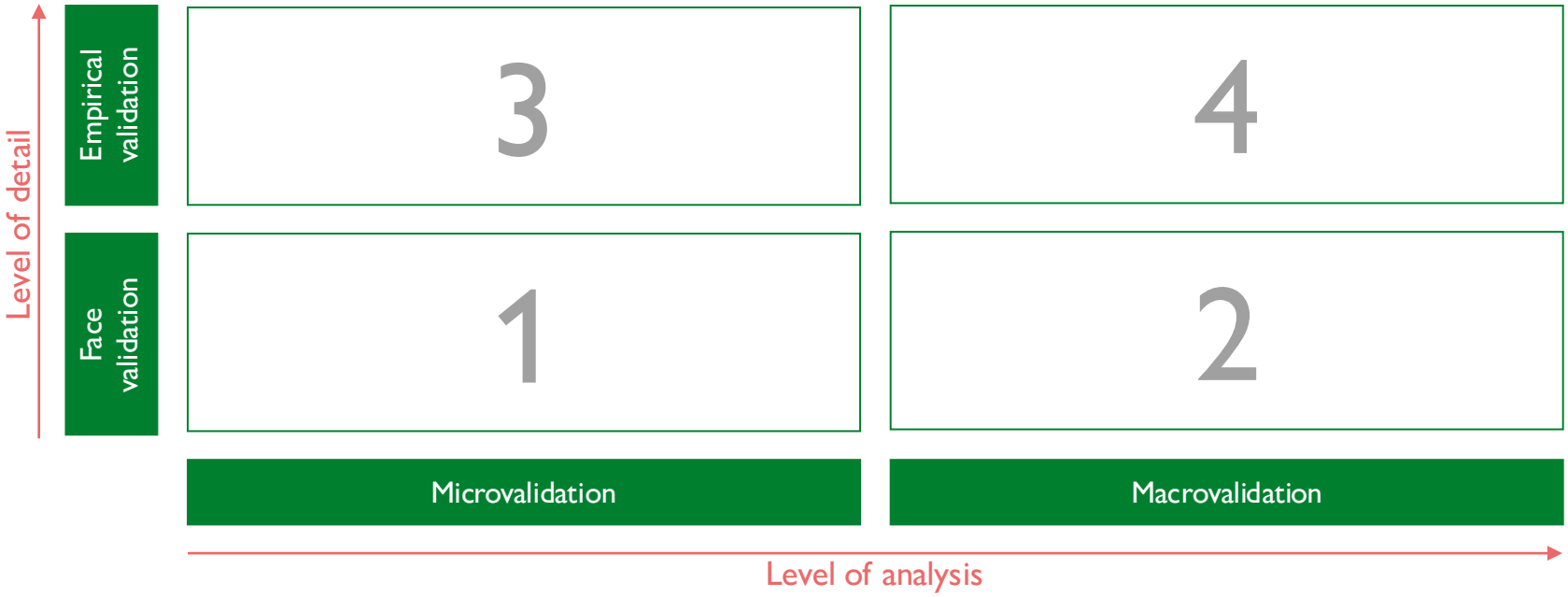
Face validation is the process of showing that the mechanisms and properties of the model look like mechanisms and properties of the real world.

Empirical validation makes sure that the model generates data that can be demonstrated to correspond to similar patterns of data in the real world."*(Wilensky, U., & Rand, W. (2015). An Introduction to Agent-Based Modeling: Modeling Natural, Social, and Engineered Complex Systems with NetLogo. The MIT Press. p. 326)*

This is the main point for making useful statements about the real world.

The result is **not binary** either!

Approaches


Level of detail
Empirical validation
Face validation
3
4
1
2
Microvalidation
Macrovalidation
Level of analysis

Own experiences

In paper reviews

- We were always challenged to provide case studies … difficult for several reasons
- We were always challenged to make abstract concepts (such as “environmental complexity”) tangible through illustrative instances

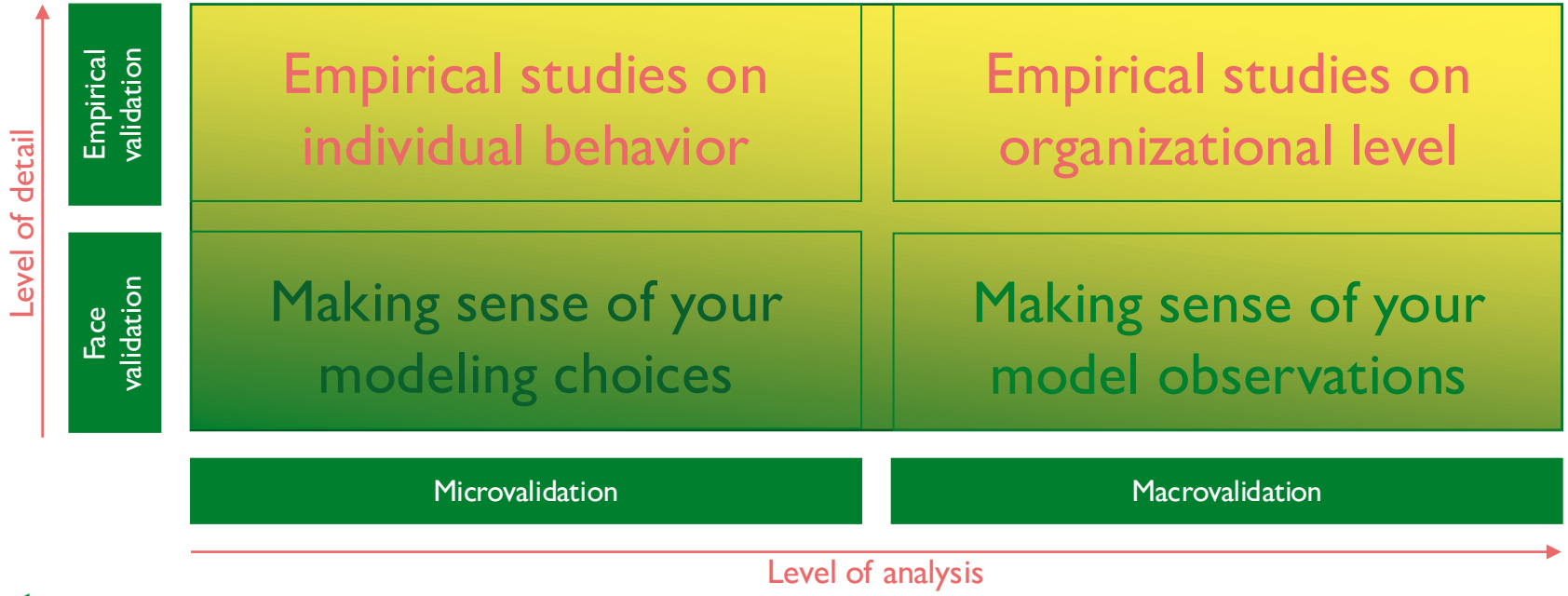
- Sometimes we were surprised how reviewers of “theoretical/theory heavy” journals argued with practical industry observations and experiences

In general

- This is really the point were the theory and practice meet

Implications for our projects: Focus on face validity

What to do in *our projects*?



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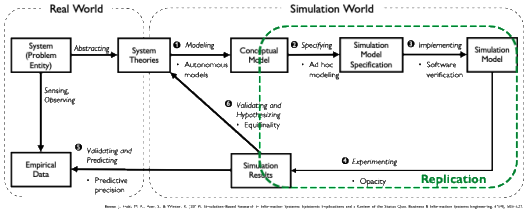

What if another researcher builds an independent model of the same phenomenon? Will the results be the same?

Generally, if results can be replicated, they become more acceptable because it decreases the likelihood that results are dependent on local, unknown conditions or stem from selective samples.

> Replication studies generally foster the understanding of a field and the maturity of research method execution. Having replication in mind often improves the reporting on the original model.

Yet, replication studies (depending on discipline) often have a problematic standing, perception, and acceptance.

Neither is this result **binary**!

Approaches

Dimensions

Dimensions in which Original and Replication may differ:

Likelihood of different results ↓

1. Time
2. Hardware
3. Languages
4. Toolkits
5. Algorithms
6. Authors

Replication Standard

How is a "successful replication" determined?

Level of difficulty to establish ↑

1. Numerical identity
2. Distributional equivalence
3. Relational alignment

Own experiences

In paper reviews

- … hardly any
- However, reviewers/editors ask to build *"baseline models"* ideally based on well-published research and to show that existing results can be replicated by the new model
- This hints at incremental developments

- Still, this is rather difficult to do.

Implications for our projects

What to do in *our projects*?

This is limited in the context of our course and project, still …

Are there simulation studies that are close to your project? What can we learn from those?

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Assignment

1. **Verification**: For your group project
    1. Is the terminology of your conceptual model consistent with the terminology used in your source code? On slides, in the code tab, interface tab, info tab?
    2. How did you interact in the group? Have there been “modelers” and “implementers”?
    3. Can you explain unexpected results on the code level?

2. **Validation**: For your group project
    1. How can you justify the agent-level behavior based on our general experience/observations?
    2. How can you justify the system-level behavior based on our general experience/observations?
    3. Is there any empirical data available that backs/contradicts your model?

3. **Replication**: For your group project
    1. What are simulation studies that are close to your’s?
    2. Can we learn something here?

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"From insight to impact"